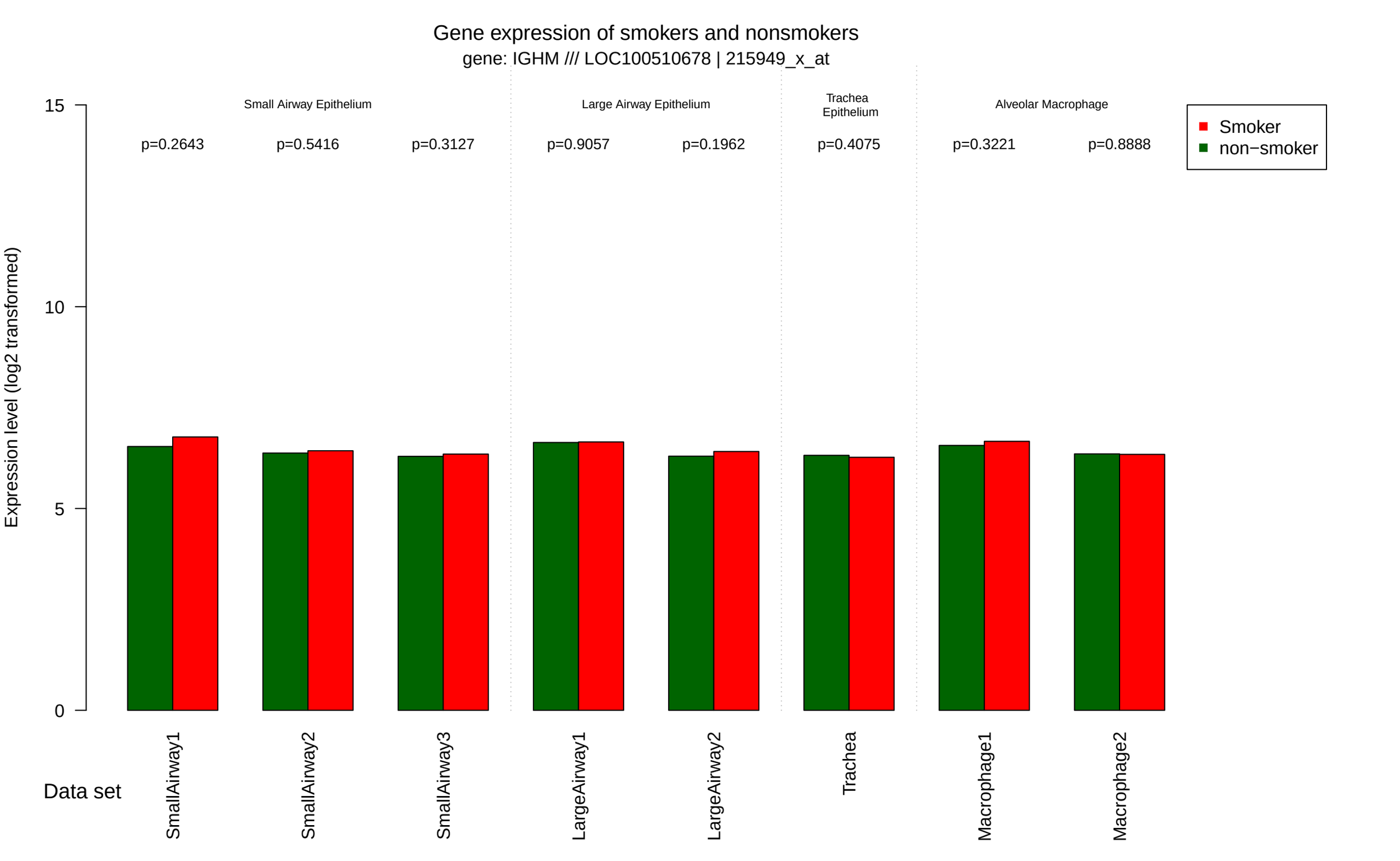
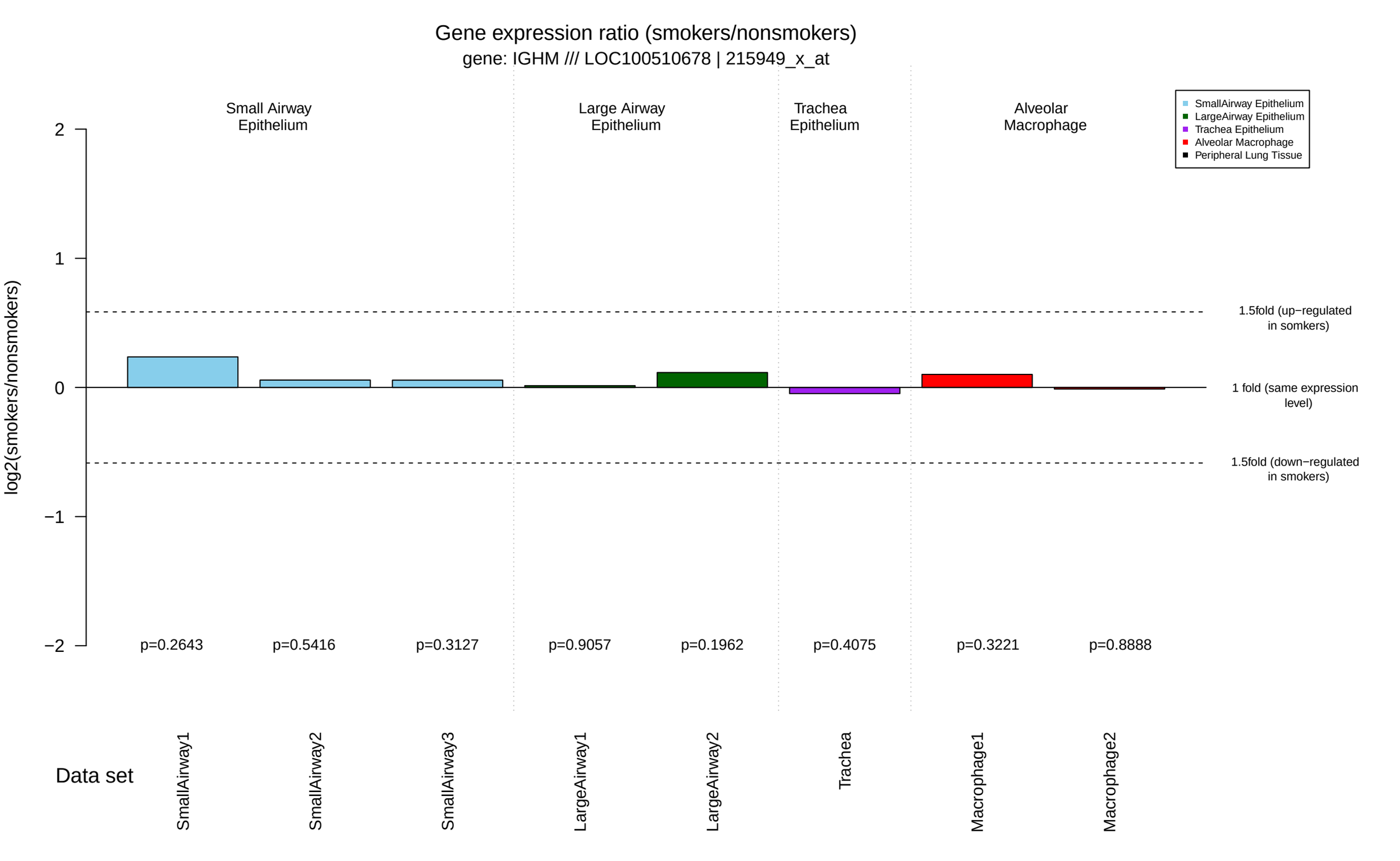
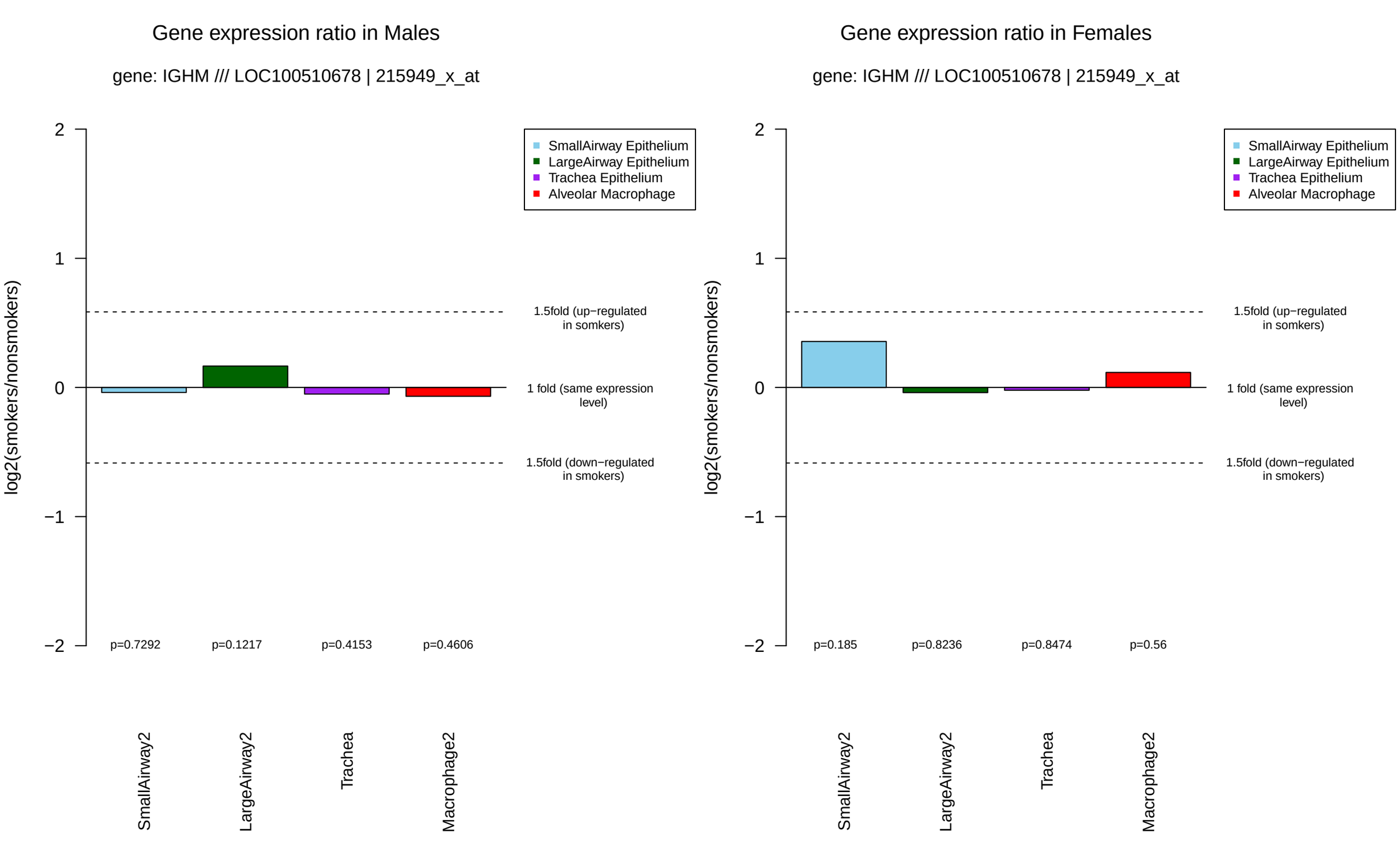
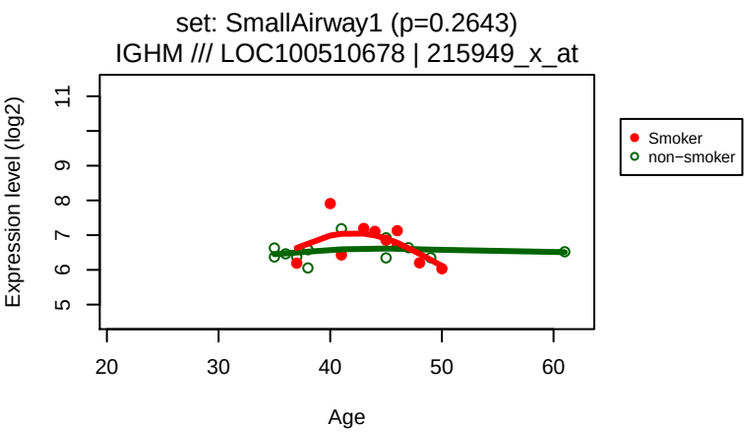
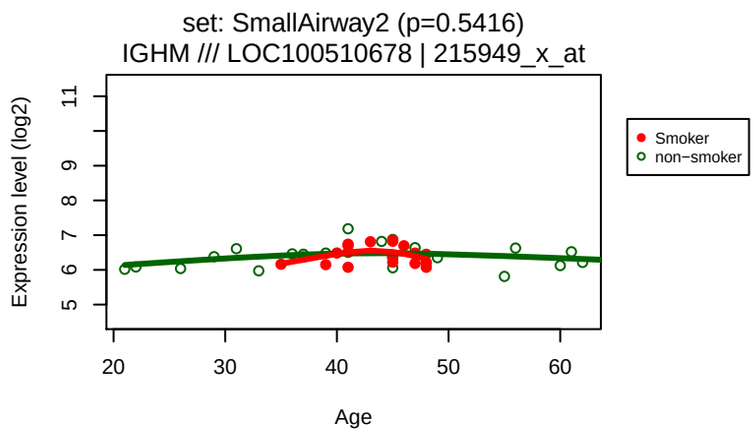
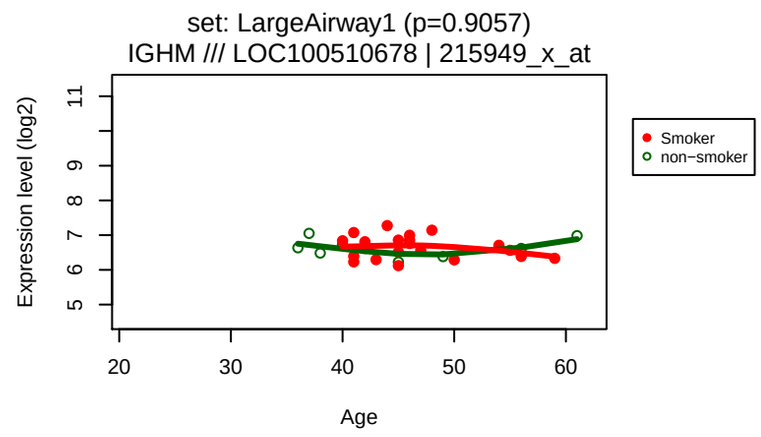

Gene expression of smokers and nonsmokers
gene: IGHM /// LOC100510678 | 215949_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2643
p=0.5416
p=0.3127
p=0.9057
p=0.1962
p=0.4075
p=0.3221
p=0.8888
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

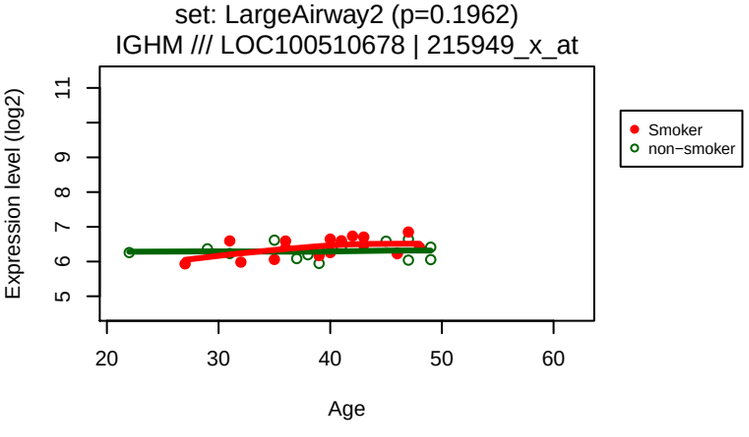

Gene expression ratio (smokers/nonsmokers)
gene: IGHM /// LOC100510678 | 215949_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2643
p=0.5416
p=0.3127
p=0.9057
p=0.1962
p=0.4075
p=0.3221
p=0.8888
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

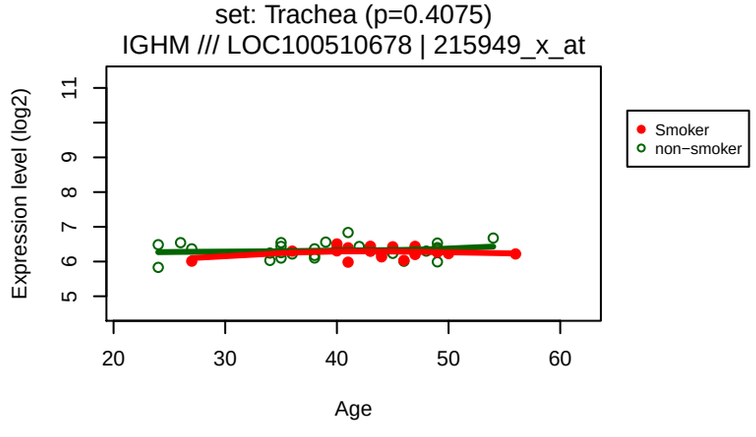

Gene expression ratio in Males
gene: IGHM /// LOC100510678 | 215949_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7292
p=0.1217
p=0.4153
p=0.4606
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHM /// LOC100510678 | 215949_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.185
p=0.8236
p=0.8474
p=0.56
SmallAirway2
LargeAirway2
Trachea
Macrophage2

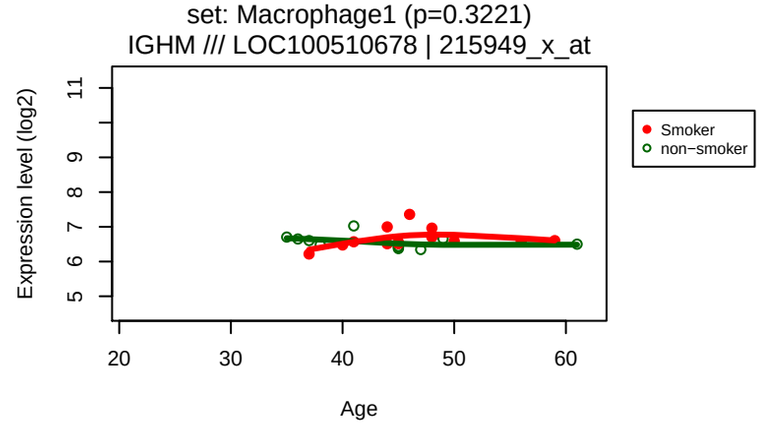

set: SmallAirway1 (p=0.2643)
IGHM /// LOC100510678 | 215949_x_at
Expression level (log2)
Age
Smoker
non−smoker

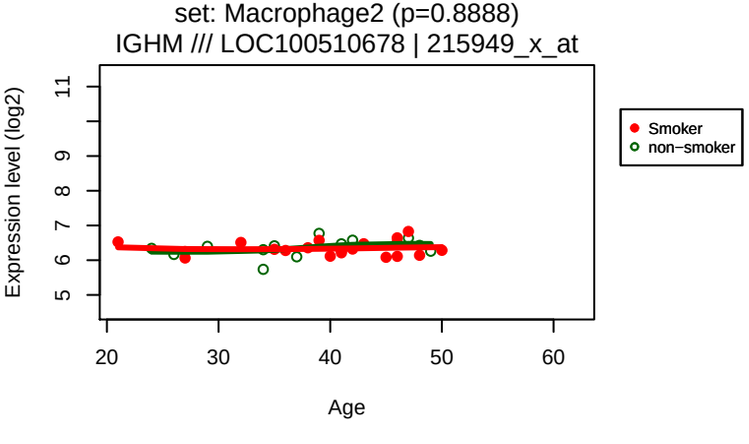

set: SmallAirway2 (p=0.5416)
IGHM /// LOC100510678 | 215949_x_at
Expression level (log2)
Age
Smoker
non−smoker

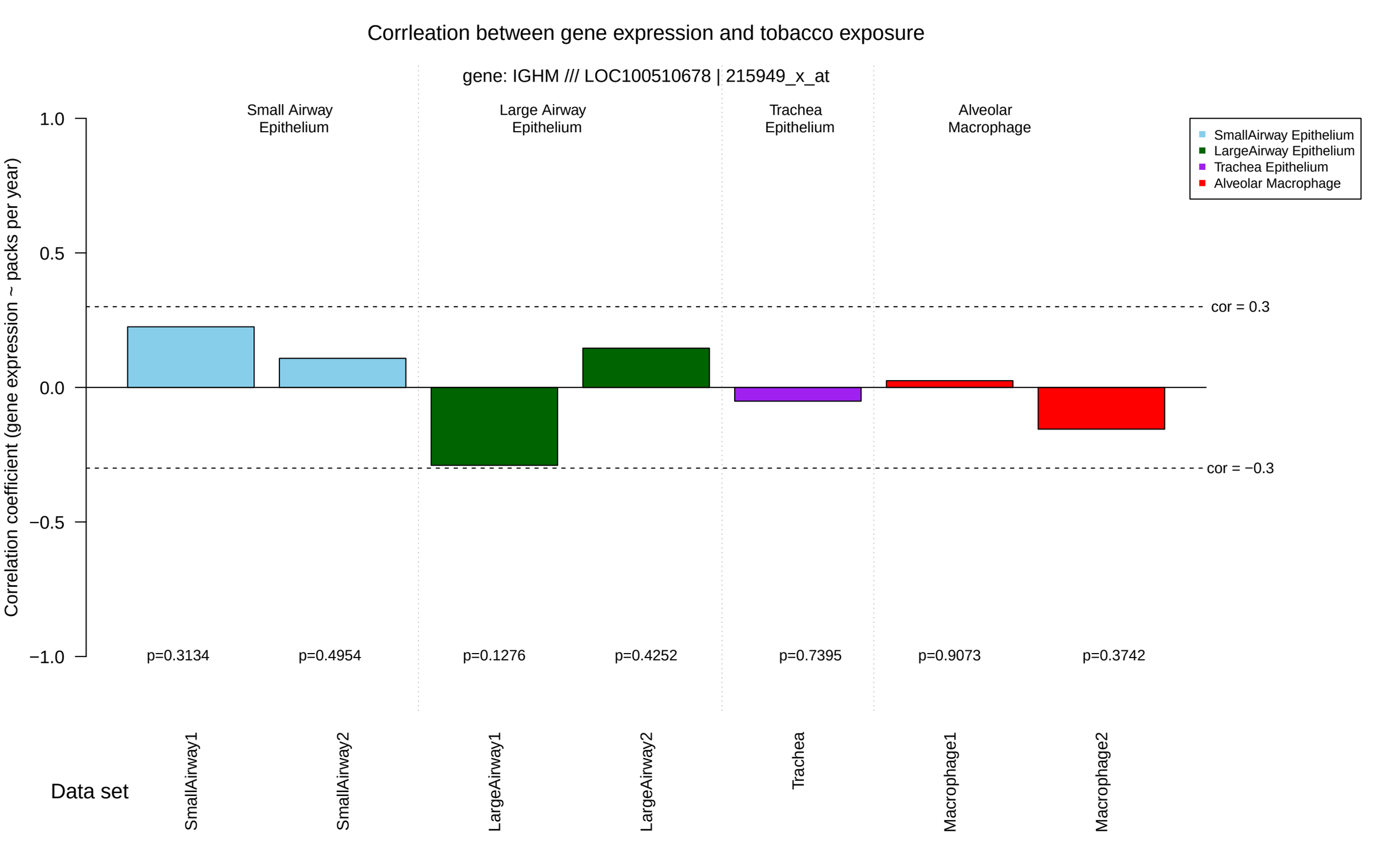

set: LargeAirway1 (p=0.9057)
IGHM /// LOC100510678 | 215949_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1962)
IGHM /// LOC100510678 | 215949_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4075)
IGHM /// LOC100510678 | 215949_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3221)
IGHM /// LOC100510678 | 215949_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8888)
IGHM /// LOC100510678 | 215949_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHM /// LOC100510678 | 215949_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3134
p=0.4954
p=0.1276
p=0.4252
p=0.7395
p=0.9073
p=0.3742
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2